EEFM Formulae

(A) Single Cash

1. $(F/P, i, n) = (1+i)^n$

2. $(P/F, i, n) = (1+i)^{-n}$
$= (F/P, i, n)^{-1}$

F → Future worth
P → Present worth
i → interest rate.
n → no. of interest periods (usually in years)

(B) Uniform Cash

1. $(F/A, i, n) = \left[\dfrac{(1+i)^n - 1}{i}\right]$

A → amount that is repeated.

2. $(A/F, i, n) = \left[\dfrac{i}{(1+i)^n - 1}\right] = (F/A, i, n)^{-1}$

3. $(P/A, i, n) = \left[\dfrac{1 - (1+i)^{-n}}{i}\right]$

OR

$= (F/A, i, n) * (P/F, i, n)$

4. $(A/P, i, n) = \left[\dfrac{i}{1 - (1+i)^{-n}}\right]$

OR

$= (F/P, i, n)(A/F, i, n)$

(C) Gradient

① $(P/G, i, n) = \left[\dfrac{(1+i)^n - in - 1}{i^2 (1+i)^n}\right]$

② $(A/G, i, n) = \left[\dfrac{(1+i)^n - in - 1}{i\left[(1+i)^n - 1\right]}\right]$

OR

$$= (P/G, i, n)\,(A/P, i, n)$$

(D) Effective Interest

☆

$$i = \left(1 + \frac{r}{CK}\right)^{C} - 1$$

effective interest rate

nominal interest rate

no. of interest periods per payment period.

number of payment periods per year

Note: Only formulae with ☆ need to be remembered, all others can be derived as a combination of the ☆ formulae.